



注解. (student-handbook)

Final Research Direction:

A valid, modular, structure-preserving surrogate workflow for fast American-option risk surface.

"如何用神经网络加速美式期权风险曲面的计算, 同时不破坏金融约束."

neural-network = an accelerator and diagnostic tool, not the backbone.

Layer 1: Classical American option Pricing. (PDE + free boundary + PSOR)

American option 有4种等价的数学描述

- <1> optimal stopping (最优停止) 选择最优时间 τ 让收益最大.
- <2> free boundary 预测 S^* (early exercise boundary)
- <3> PDE (continuation region)
- <4> Variational Inequality 不能用PDE, 还要满足价值 $\geq$ 立即行权价值.

Layer 2: Scientific Machine Learning for PDEs (PINN + DGM + SciML).

PDE { PSOR 求 PDE 的数值解
神经网络学 PSOR 解的映射条件

PINN (Physics-Informed Neural Network)

· 普通神经网络: 输入 $x \rightarrow$ 神经网络 $\rightarrow$ 输出 y (需要大量标签).

· PINN 直接学习 PDE 的解.

{ 普通方法: 有限差分 $\rightarrow$ 得到很多 $(u, x, t) \rightarrow$ 训练网络

{ PINN: 神经网络直接输出 $u(x, t)$, 然后自动求导 u_t, u_x, u_{xx}
把 PDE 残差放进 loss. $Loss = data\ loss + PDE\ residual$

若 PDE 满足 $Residual \approx 0$
认为学到了 PDE 解.

DGM (deep galerkin Method)

随机采样, 直接学习 PDE 解. (适合高维)

Note!!! American Option 有 obstacle constraint.

Layer 3: Deep learning for American Options.

optimal stopping methods.
tree boundary and PINN methods.

focus on risk surfaces, boundary recovery, Greek stability and financial consistency.

Layer 4: Application Direction

Mathematical Formulation

1. Risk-Neutral Model

$$dS_t = (r - q) S_t dt + \sigma S_t dW_t \quad (1)$$

$$V(S, t) = \sup_{\tau \in [t, T]} \underbrace{\mathbb{E}}_{\text{expectation}} \left[\underbrace{e^{-(r-q)(T-t)}}_{\text{discount (折现)}} \Phi(S_t) \mid S_t = S \right] \quad (2)$$

取最大值

for a put and a call

$$\Phi^{\text{put}}(S) = (K - S)^+ \quad \Phi^{\text{call}}(S) = (S - K)^+ \quad (3)$$

dW_t : 布朗运动

S_t : underlying asset price

r : risk-free rate

q : continuous dividend yield 股息率

σ : volatility 波动率

maturity T

payoff $\Phi(S)$

2. Variational Inequality

$V_s = \text{Delta}$

$V_{ss} = \text{Gamma}$

Black-Scholes Operator:
$$\mathcal{L}V = \frac{1}{2}\sigma^2 S^2 V_{ss} + (r-q)SV_s - rV \quad (4)$$

In original time t , the American option satisfies the complementarity conditions.

$$\star \begin{cases} V(s,t) \geq \bar{X}(S) & (5) \rightarrow \text{期权价值永远不能低于立即行权收益} \\ -(V_t + \mathcal{L}V) \geq 0 & (6) \rightarrow \text{继续持有区域 满足 BS PDE} \\ (V(s,t) - \bar{X}(S))[-(V_t + \mathcal{L}V)] = 0 & (7) \rightarrow \text{互补条件} \end{cases}$$

In the continuation region, $V > \bar{X}$ and the Black-Scholes PDE holds.

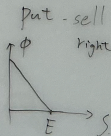
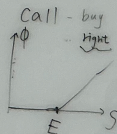
In the exercise region, $V = \bar{X}$

6.16

S (stock price)

Underlying asset $\Leftarrow$ derivatives hedging
Greeks

$$\Delta = \frac{\partial P}{\partial S}, \text{ Gamma } \frac{\partial^2 P}{\partial S^2}, \left(\text{e.g. } \frac{\partial P}{\partial r}, \text{ Vega } \frac{\partial P}{\partial \sigma}, \text{ then } \frac{\partial^2 P}{\partial \sigma^2} \right)$$



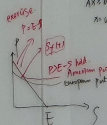
premium
||
option price (V)

BS (Black-Scholes) $\frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$

European: $V(S, T) = \Phi(S)$ (Terminal)

Call: $V(S, T) = \max(S - K, 0)$
Put: $V(S, T) = \max(K - S, 0)$

American: $V(S, t) \geq \Phi(S)$
 $V(S, t) = \sup_{\tau \in [t, T]} E^Q [e^{-r(\tau-t)} \Phi(S_\tau)]$



LCP $\Rightarrow$ FDM $\Rightarrow$ P3OR

General:

$$AX = b, X \geq 0, \text{ and } A \geq 0$$

$$A = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}$$

$$b = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

$$X = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}$$

$$V = \max(S, K) \text{ for } t \leq T \Rightarrow \frac{\partial V}{\partial S} = 1$$

$$V = \max(K, S) \text{ for } t \leq T \Rightarrow \frac{\partial V}{\partial S} = -1$$

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2026.06.16 (学姐).

option (衍生品的一个类别) 非线性问题, 需要不断 hedging.

Greeks

underlying asset $\in$ derivatives hedging
 S (stock price)

$$\text{delta } \Delta = \frac{\partial P}{\partial S}, \quad \text{Gamma } \frac{\partial^2 P}{\partial S^2}$$

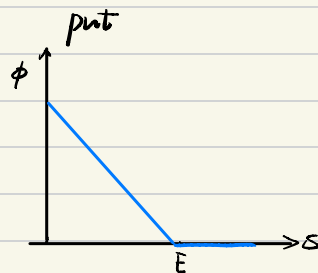
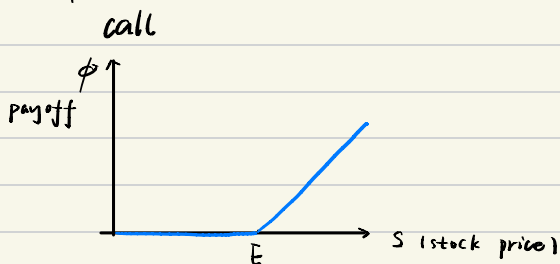
$$\text{eg. rho } \frac{\partial P}{\partial r}, \quad \text{vega } \frac{\partial P}{\partial \sigma}, \quad \text{theta } \frac{\partial P}{\partial T}$$

r : interest rate

波动率

P : portfolio

σ : standard deviation



call / put : 买和卖出的权力

给权力定价

premium

||

option price

BS (heat-equation)

$$V_t + \frac{1}{2}\sigma^2 S^2 V_{SS} + rSV_S - rV = 0$$

European option : $V(S, T) = \phi(S)$ (terminal condition)

SS 没有 dividend yield.

call

$$V(0, t) = 0$$

American call

$\Downarrow$
C.

$$V(S_{\max}, t) = S_{\max} - \underbrace{ke^{-r(T-t)}}_{\text{discount factor}}$$

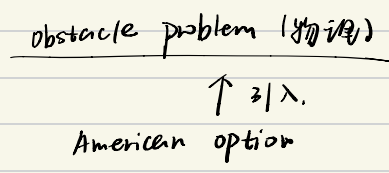
continuous

解析解.

!!!

American option
put.

$$V(s, t) = \sup_{\tau \in [t, T]} E^Q \left[e^{-r(\tau-t)} \bar{z}(S_\tau) \right]$$



$\Rightarrow$ PDM $\xrightarrow[\text{迭代}]{\text{OT}}$ PSOR

$$t_n = \frac{T}{N}, \quad t_n = n \cdot t, \quad n = 0, 1, \dots, N$$

$$V_i^n = V(s_t, t_n) \quad \text{solve } t=T \Rightarrow t=0.$$
$$j = 1, \dots, M-1 \quad V_S(\text{Sit}_n) = \frac{V_{n+1}^n - V_{i-1}^n}{2\sigma_S}$$

(内部为点).

$$V_i^n - V_i^{n+1} = \theta (L_h V^n)_i + (1-\theta) (L_h V^{n+1})_i \quad // \text{从后往前推的过程}$$

从后往前推的过程

direct update



✓ 6-16 知识清单



✓ Greeks



✓ payoff function



✓ Black-Scholes PDE

{
PDE
CN
PINN
DGM



✓ American option vs European option



Linear complementarity problem, FDM



PDE 离散化变成 $Ax=b$ 求解

CN

option basics → call & put → Greeks → Black-Scholes PDE

→ partial derivatives → discretization → Finite Difference Method → CN → American option

Option Basics.

option 是一种金融衍生品, 它给买方一个权利, 而不是义务, 在未来某个时间, 以事先约定的价格买入或卖出标的资产.

option buyer has the right, but not the obligation.

option 的五个核心要素

- | | | |
|------------|-------------------|----------------------------------|
| ① S | underlying price | (当前标的的价格) |
| ② K | strike price | 行权价 (fixed when contract create) |
| ③ T | Maturity / expiry | 到期时间 |
| ④ r | Risk-free rate | 无风险利率 |
| ⑤ σ | volatility | 波动率 |

option premium

期权价格 (买期权要付的钱)

常见标的资产

股票: apple stock

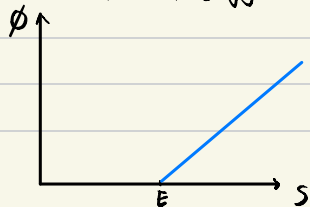
指数: S & P 500

外汇

商品: Gold, oil

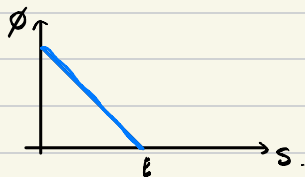
利率: bond option

Call 到期 payoff $\max(S_T - K, 0)$



给买方一个权利, 在未来以固定价格 K 买入 underlying asset.

put 到期 payoff $\max(k - S_T, 0)$



给买方一个权利, 在未来以固定价格 k 卖出 underlying asset.

In the money

At the Money

Out of the Money

有行权价值

刚好平价

不值得行权

$$S = k \begin{cases} \text{call} : S > k \\ \text{put} : S < k \end{cases}$$

$$\begin{cases} \text{call} : S < k \\ \text{put} : S > k \end{cases}$$

对比	European Option	American Option
行权时间	只能到期行权	到期前任意时间可行权
灵活性	低	高
定价难度	较简单	更难
是否有 closed-form	Black-Scholes 有欧式公式	一般没有解析解
常用方法	Black-Scholes formula, PDE, Monte Carlo	Binomial tree, finite difference, LSM, PDE
是否有 early exercise boundary	没有	有
项目重点	baseline	研究重点

American option & European option

★ American put:

$$V(s, t) = \max(\text{continuation value}, k - S)$$

American option $\rightarrow$ Early exercise boundary 尤其是 American put. 寻找临界 $S^*(t)$

American call: 如果 underlying stock 不支付股息, 通常不提前行权
 $\therefore$ 对于 non-dividend-paying stock: $C_{\text{American}} = C_{\text{European}}$
American put

option pricing 在风险中性世界下, 计算未来 payoff 的折现期望.

European option price: $V(S,t) = e^{-r(T-t)} E^Q [\text{payoff}]$ r : 无风险利率

BS PDE:
$$\frac{\partial V}{\partial t} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

描述期权价格如何随时间、underlying asset, volatility and interest rate 变化

European option 的 terminal condition

• European call: $V(S,T) = \max(S-T, 0)$

• European put: $V(S,T) = \max(K-S, 0)$

PDE 从 maturity 的 payoff 往前倒推.

American option
★ 满足 PDE
难点 inequality constraint
 $V(S,t) \geq K-S$

Premium

买期权时付的钱

payoff — call: $\max(S_T - K, 0)$
— put: $\max(K - S_T, 0)$

到期时, 这张期权
本身值多少钱
不考虑最开始付的钱

profit

真正赚的钱
减去 premium

call: 买入的权利

put: 卖出的权利

long / short 决定买还是卖

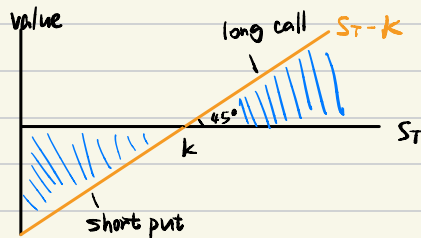
put-call parity 严格成立的前提是 European Option

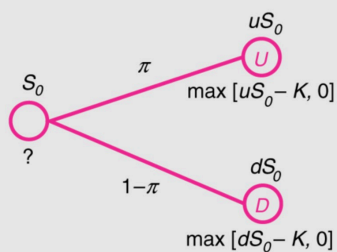
$$\text{即 } C_E - P_E = S - PV(K)$$

Position	含义
Long	买入
Short	卖出
Long Call	买入看涨期权
Short Call	卖出看涨期权
Long Put	买入看跌期权
Short Put	卖出看跌期权

European Options: put-call parity

American Options: { call { with dividend pay
without dividend pay
put options }





Valuing options in isolation $\leftarrow$ 未来payoff + Risk-Neutral Pricing

π = Risk-Neutral Probability.

满足 $S_0 = \frac{1}{R} [\pi Su + (1-\pi)Sd]$

在这种虚拟概率下, 股票的预期收益率刚好等于无风险利率.

Risk-neutral probabilities

Different ways of looking at it.

- ① "Mathematical Trick" that avoids calculating the portfolio replication strategy...
- ② We know from chapter 9:

Consumption-Based Asset Pricing

$$P_i = \frac{E[U'(X^*)X_i]}{E[U'(X^*)]R_f}$$

For two outcomes (up, down) this is:

$$\begin{aligned} P_i &= Pr(up) \cdot \frac{U'(X_{up}^*)X_{i,up}}{E[U'(X^*)]R_f} + Pr(down) \cdot \frac{U'(X_{down}^*)X_{i,down}}{E[U'(X^*)]R_f} \\ &= \frac{(\pi_{up}X_{i,up} + \pi_{down}X_{i,down})}{R_f}, \end{aligned}$$

$$\text{with } \pi_{up} = Pr(up) \cdot \frac{U'(X_{up}^*)}{E[U'(X^*)]}.$$

资产价格取决于:

未来赚多少钱 + 赚钱在什么时候.

二叉数 $N \rightarrow \infty$ 得到 BS Model.

Greeks = 期权价格对各种因素的敏感度.

$$V = V(S, t, \sigma, r)$$

Delta (Δ) $\Delta = \frac{\partial V}{\partial S}$

stock price 影响 option value

价格 - 阶导

Gamma (Γ) $\Gamma = \frac{\partial^2 V}{\partial S^2}$

delta 变化的速度.

二阶

Theta (θ) $\theta = \frac{\partial V}{\partial t}$

时间过去一天, 期权损失多少钱

时间 - 阶导.

Vega (ν) $\nu = \frac{\partial V}{\partial \sigma}$

波动率 增加 1%, 期权涨多少?

Rho (ρ) $\rho = \frac{\partial V}{\partial r}$